GA-MAE: Gradient-Guided Activation-Aware Masked Autoencoder for Remote Sensing Image Classification

Xiangyu Xu, Zhixi Feng, *Member, IEEE*, Shuyuan Yang, *Senior Member, IEEE*, Mengchang Li, and Gechang Yao

***Abstract*—In the field of remote sensing self-supervised learning, the self-supervised learning paradigm based on masked image modeling (MIM) effectively promotes the learning of structural and contextual semantic information in images. However, the random masking strategies employed in past studies have not effectively utilized the distribution of semantic information within images. Furthermore, overly simple reconstruction tasks cannot effectively improve self-supervised learning performance. To address these challenges, this article introduces a gradient-guided activation-aware autoencoder (GA-MAE). Specifically, this approach computes and captures the spatial spectral activation map (SSAM) with semantic distribution information during the process of backpropagation of spatial–spectral reconstruction loss (SSR Loss) in the pretraining phase, thereby enabling masking operations on similar visual patches with higher reconstruction loss. Additionally, we introduce an activation-aware self-attention mechanism (ASA) that adjusts the self-attention dependencies by utilizing the weight information of visual patches provided by SSAM. Extensive experiments conducted on three public datasets (Houston2013, Augsburg, and Berlin) demonstrate the effectiveness of the proposed approach. With only ten labeled training samples per class, the proposed method achieves an overall accuracy (OA) of 91.24%, 84.69%, and 73.43% on the Houston2013, Augsburg, and Berlin datasets, respectively, indicating improvements of 1.09%, 2.05%, and 2.40% over state-of-the-art methods.**

***Index Terms*—Few-shot classification, gradient-guided masking, masked image modeling (MIM), multimodal data, self-supervised learning (SSL).**

I. Introduction

THE groundbreaking advancements in remote sensing technology have fundamentally redefined Earth observation methodologies [1]. By integrating multimodal sensor data, including hyperspectral imaging (HSI) with rich spectral signatures, Light Detection and Ranging (LiDAR) providing 3-D elevation information, and synthetic aperture radar (SAR) enabling all-weather geometric observation, modern remote sensing systems have established a multidimensional Earth observation framework [2]. This paradigm shift enables all-weather, multiscale data acquisition, providing critical support for precision agriculture [3], urban planning [4], geohazard monitoring [5], [6], [7] and emergency management [8]. In multimodal remote sensing analytics, the cross-modal fusion of heterogeneous sensor data has emerged as a cornerstone for improving interpretation accuracy. The 3-D elevation information from LiDAR can be synergistically integrated with HSI-derived spectral signatures [9], [10], [11], facilitating stratified identification of ground objects across vertical profiles. Concurrently, the joint analysis of hyperspectral data (with spectral discrimination) and SAR data (with geometric robustness) leverages complementary geometric deformation compensation through spectral-radar feature fusion [12], [13], [14]. Such multimodal collaborative frameworks not only expand the dimensionality of feature spaces but also enhance model robustness via cross-validation of heterogeneous data sources, thereby reducing dependency on single-modality inputs. However, the heterogeneity of multimodal data (HSI, LiDAR, SAR) with region-specific semantic distributions brings unique challenges: traditional random masking and static attention fail to handle cross-modal semantic weight nonuniformity and inter-modal complementarity complexity.

Due to the scarcity of high-quality annotated samples [15], current research paradigms typically integrate self-supervised learning with few-shot learning [16]. Self-supervised learning extracts discriminative features from unlabeled data via unsupervised pretraining, reducing reliance on annotations and enabling models to capture spatial-spectral patterns for efficient few-shot adaptation [17], [18].

Self-supervised learning frameworks in remote sensing primarily follow two pathways: contrastive learning [19] and generative learning [20]. Contrastive learning optimizes cross-modal representation alignment via positive-negative pair matching [21]. Masked image modeling (MIM), a representative generative approach, trains models to learn contextual correlations of geographic elements by reconstructing randomly masked image regions [22], [23]. Pretraining on large unlabeled datasets demonstrates that MIM achieves superior generalization across downstream tasks such as land-use classification and object detection [24]. MIM excels by capturing global layouts and local details via hierarchical encoding

Received 27 April 2025; revised 29 July 2025 and 20 October 2025; accepted 20 November 2025. Date of publication 25 November 2025; date of current version 8 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62276205 and Grant U22B2018, in part by China Postdoctoral Science Foundation under Grant 2024M762556, and in part by the Fundamental Research Funds for the Central Universities under Grant XJSJ24069. *(Corresponding author: Zhixi Feng.)*

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Digital Object Identifier 10.1109/TGRS.2025.3637094

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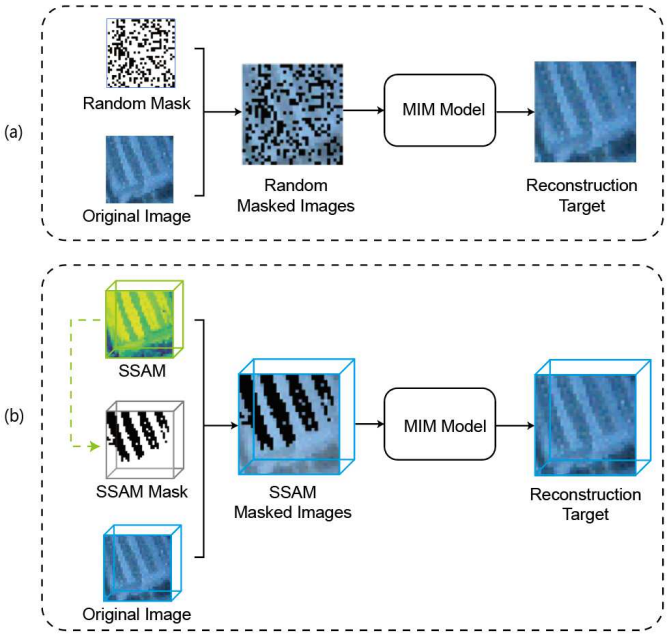

Fig. 1. Comparison between GAM and random masking. (a) Traditional MIM random masking. (b) SSAM-guided targeted masking.

and pixel-level decoding, enhancing remote sensing scenario understanding [25].

Despite the excellent advancements of self-supervised-based MRSI classification methods mentioned above, there are still two issues.

1 *Random Mask:* As shown in Fig. 1, the conventional random masking strategy, based on the assumption of uniform probability, randomly masks image patches [20], [23], [26]. However, this contradicts the nonuniform distribution of semantic information in images. In multimodal scenarios, this is worsened by modality-specific semantic contributions (e.g., HSI spectral details versus LiDAR structural features dominating different regions), causing random masking to miss cross-modal critical patches [27]. Moreover, overly simple proxy tasks allow models to easily leverage contextual information for reconstruction, resulting in insufficient learning capacity. Existing attention-based masking methods struggle to identify critical regions in high-resolution remote sensing due to blurred foreground-background boundaries [28], [29].

2 *Spatial and Channel Attention Limitations:* While spatial attention mechanisms focus on pixel-level positional relationships, they overlook the discriminability of spectral bands, thereby weakening cross-band feature representation and the extraction of subtle spectral features [30], [31]. Channel attention with global pooling loses structural details during spatial compression, particularly at region boundaries [32]. Although hybrid methods integrate both spatial and channel attention, decoupled weight computation lacks dynamic spectral-spatial interaction, failing to jointly model spectral-spatial features with pixel-level precision [33]. In multimodal processing, this limitation is amplified: existing attention cannot dynamically balance heterogeneous features (e.g., HSI bands versus LiDAR elevation) or capture context-dependent interactions, resulting in suboptimal cross-modal fusion.

To address the aforementioned limitations, this study proposes a gradient-guided activation-aware autoencoder pretraining framework (GA-MAE). The framework significantly enhances remote sensing image classification performance through the collaborative optimization of spatial–spectral adaptive masking and dynamic attention modulation. Tailored for multimodal inputs, GA-MAE uses gradient-guided activation-aware masking (GAM) to capture cross-modal semantics via spatial spectral activation map (SSAM) (masking intermodal hard regions) and activation-aware self-attention mechanism (ASA) to adjust attention via SSAM weights, enhancing multimodal complementary modeling. During the pretraining phase, a gradient-aware masking module is designed to construct more challenging proxy tasks. GAM calculates and captures the SSAM, which encodes semantic distribution information, during the backpropagation of spatial–spectral reconstruction loss (SSR Loss), and generates targeted spatial and spectral masks accordingly. Simultaneously, the SSAM is dynamically integrated into the key–value pair generation process of Transformer encoders and decoders, forming the ASA mechanism. In multihead attention computation, query vectors interact with SSAM-modulated key–value pairs, enabling the model to focus on highly discriminative regions in both spatial and spectral dimensions. Ultimately, long-range dependency modeling for challenging samples is optimized through dual-dimensional attention focusing on the spatial–spectral domain. The main contributions are summarized as follows.

1 A novel GAM strategy is proposed. This strategy can precisely identify regions that are difficult for the reconstruction task, overcome the limitations of traditional random masking, and enhance the performance of MIM in downstream remote sensing tasks.
2 An activation-aware self-attention (ASA) strategy is introduced. By enhancing the feature responses of difficult regions in multihead attention, it enables dynamic spatial–spectral interactions and optimizes the modeling ability for difficult spatial and spectral information in remote sensing images.
3 A gradient-guided activation-aware autoencoder framework (GA-MAE) is developed. Through collaborative optimization of spatial–spectral adaptive masking and dynamic attention modulation, it significantly improves the classification performance of difficult samples and boosts the efficiency of self-supervised learning.

II. Related Work

A. Mask Image Modeling

The development of self-supervised MIM traces back to the successful adaptation of masked language modeling (MLM) principles from natural language processing to computer vision. Early visual self-supervised attempts, such as iGPT

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[34], sought to model image structures through autoregressive pixel prediction, yet their effectiveness was constrained by inefficient sequence modeling frameworks. This paradigm underwent a transformative shift with the introduction of the Vision Transformer (ViT) [35], which demonstrated the efficacy of patch-based image tokenization and established the architectural blueprint for modern MIM frameworks.

Bao et al. [26] pioneered discrete token reconstruction in visual domains by leveraging a pretrained discrete variational autoencoder (dVAE) to enable semantic prediction of masked regions, marking a critical step toward semantic-level reconstruction. Concurrently, He et al. [20] introduced an asymmetric encoder–decoder architecture with a 75% high masking ratio, establishing a dual-track reconstruction paradigm that unifies pixel-level intensity restoration and feature-level semantic reconstruction. Subsequent advancements focused on optimizing model efficiency and reconstruction fidelity: Xie et al. [23] streamlined symmetric encoder–decoder designs and validated the practicality of normalized pixel regression; Chen et al. [36] enhanced discriminative representation learning through task-decoupled feature extraction and reconstruction; while Zhou et al. [37] and Liu et al. [38] improved masked region consistency by integrating knowledge distillation into the training pipeline.

As methodologies matured, MIM has been adapted to remote sensing applications with notable innovations. Lin et al. [39] proposed a generative self-supervised learning framework tailored for multimodal remote sensing data, laying the groundwork for cross-modal feature learning. SatMAE [40] further addressed temporal and multispectral characteristics of satellite imagery by introducing temporal embeddings, independent masking across time, and spectral positional encodings for grouped bands, achieving significant gains in land cover classification and semantic segmentation.

B. Mask Strategy

Inspired by MLM in natural language processing, early MIM approaches adopted random masking strategies [20], [23], [26]. Taking MAE as an example, it randomly obscures up to 75% of image patches (retaining only sparse visible patches for encoder input), compelling the model to reconstruct missing regions through global contextual reasoning. While this strategy leverages randomness to ensure diversified feature learning, it overlooks intrinsic semantic correlations and spatial structures in images, resulting in insufficient targeted guidance for modeling critical regions.

Liu et al. [28] dynamically generates semantic saliency heatmaps using the self-attention mechanism in ViTs, prioritizing the masking of high-response regions (e.g., object entities) to enhance the model's focus on discriminative features. Zhan et al. [29] aggregates cross-head attention maps of the [CLS] token in ViT encoders to produce averaged attention maps, subsequently selecting high-response regions as mask candidates to concentrate masking on semantically pivotal content.

However, attention maps inherently have a limitation for reconstruction-oriented tasks: they primarily reflect the statistical correlation between features, yet such high responses stem solely from the statistical properties of the data itself and lack direct alignment with the reconstruction task's demand for "hard-to-reconstruct regions." For instance, in SAR images, grayscale jumps caused by noise may be misidentified as high-correlation regions [41], which nonetheless offer no practical value for reducing reconstruction errors.

In remote sensing, mask strategies have been tailored to data specifics: CSMAE [42] designed three multimodal masking correspondences (identical, random, and disjoint) to control cross-sensor reconstruction difficulty. FG-MAE [43] combined spatial (HOG) and spectral (NDI) feature reconstruction, implicitly guiding masks toward semantically informative regions. These strategies address the nonuniform semantic distribution in remote sensing imagery, improving the efficiency of self-supervised learning.

III. Proposed Method

A. Overview

In this section, we introduce the details of our proposed GA-MAE framework, which consists of the GAM module, a dual-branch MIM model, and the ASA. In our approach, the GAM module captures gradient-activated regions during backpropagation of the reconstruction loss to generate SSAM that carry discriminative information. SSAM is then used in conjunction with spatial and spectral masking strategies to construct the proxy task for MIM. Simultaneously, the SSAM is dynamically integrated into the key–value pair generation process of the Transformer encoder and decoder, thereby introducing the ASA. To integrate multimodal heterogeneous features containing both spatial and spectral information, the model performs feature-level fusion followed by reconstruction. The proposed method is illustrated in Fig. 2.

B. GAM Module

As shown in Fig. 3, the proxy tasks in MIM based on random masking are insufficiently challenging, limiting the performance gains of self-supervised learning. Therefore, constructing a more challenging proxy task is essential. To this end, we designed a gradient-aware masking module to predict the importance weights of the input data $I \in \mathbb{R}^{H\times W\times C}$ with respect to the reconstruction loss. As depicted in Fig. 3, the backbone architecture of the GAM module is derived by freezing the weights of the encoder (θe), decoder (θd), and the final reconstruction convolution layer (θc) of the main model

$$\mathcal{M}_{\text{GAM}} = \mathcal{M}_{\text{main}}\left(f_r\left(\theta e, \theta_d, \theta_c\right)\right) \tag{1}$$

where $\mathcal{M}_{\text{GAM}}$ represents the architecture of the GAM module, $\mathcal{M}_{\text{main}}$ denotes the architecture of the main model, and f_r indicates the freezing operation.

In each iteration, the input data is organized as patches, denoted as $I \in \mathbb{R}^{B\times C\times H\times W}$, where B represents the batch size, C is the number of channels, and $H \times W$ indicates the resolution of each patch. We first concatenate the multimodal data and use it as the input to the GAM module. As depicted in Fig. 3, the SSR Loss used in this study aligns with that employed in the pretraining framework, employing the conventional MIM reconstruction loss. This approach ensures

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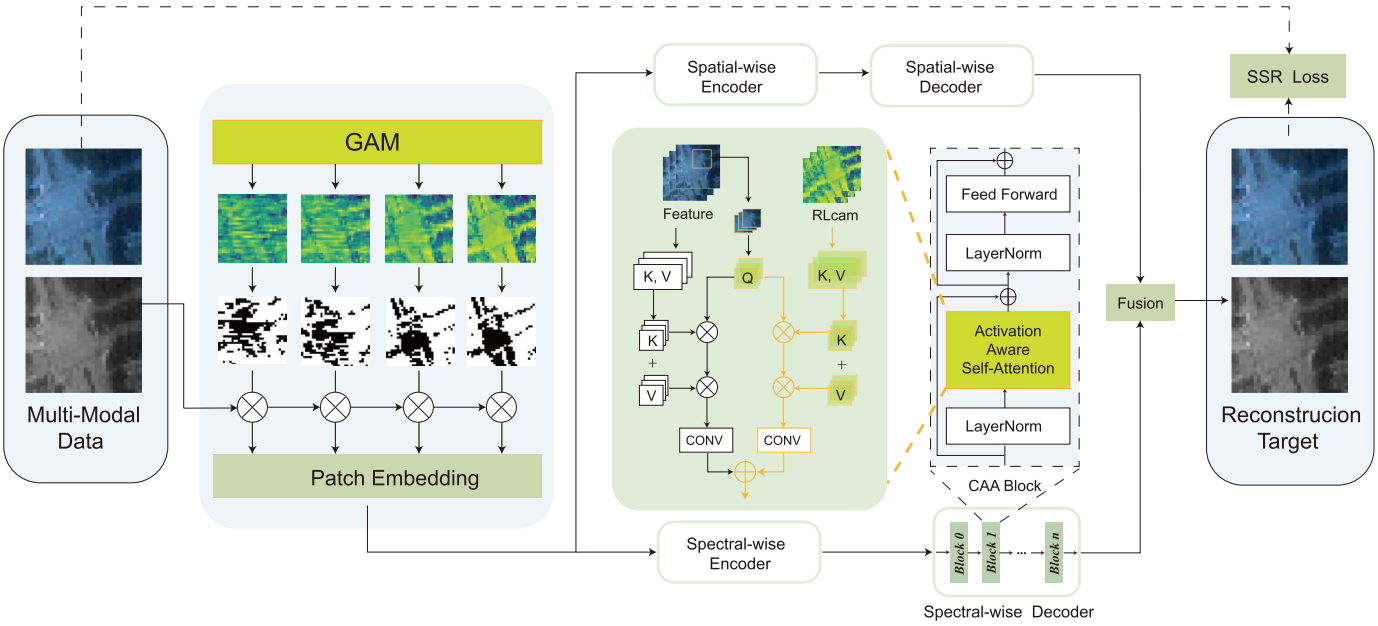

Fig. 2. Schematic of the gradient-guided activation-aware masked autoencoder (GA-MAE). The framework integrates two key components: 1) the GAM module, which adaptively generates the SSAM to mask regions/bands with high reconstruction loss in multimodal inputs, thereby optimizing masking strategies to enhance model learning capacity and 2) the ASA mechanism designed in the Transformer-based autoencoder dynamically adjusts attention weights through the SSAM, enhancing the modeling capability of feature dependencies for challenging spatial regions and spectral dimensions.

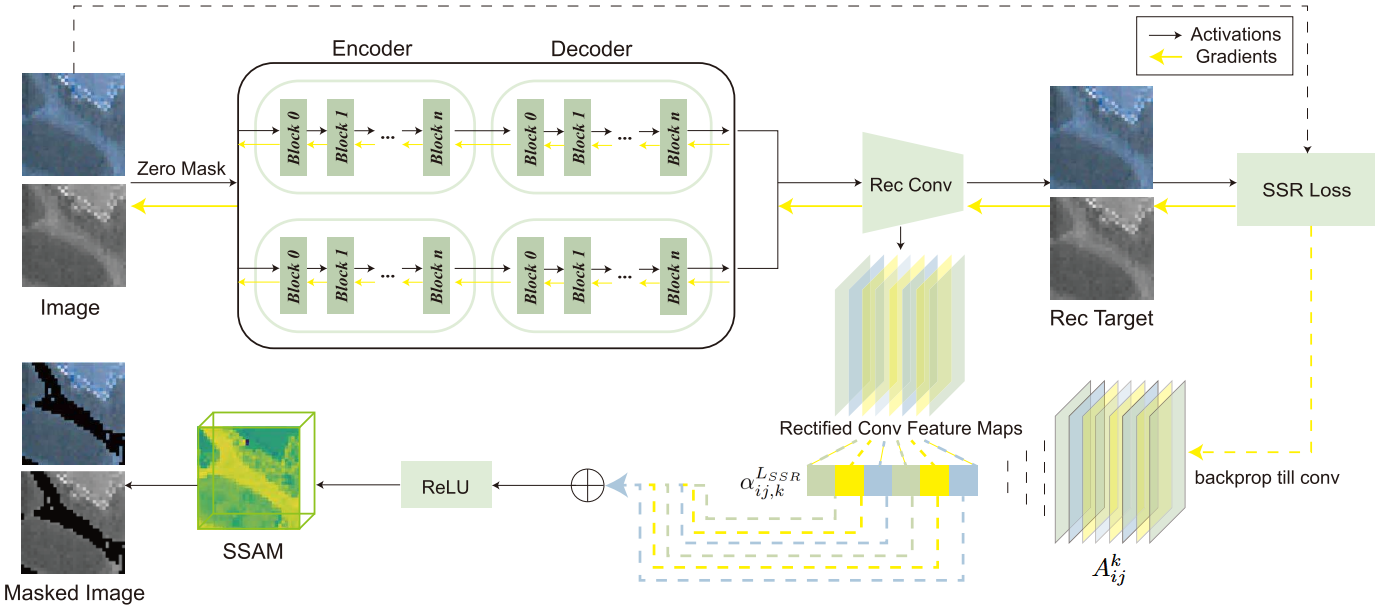

Fig. 3. Overall framework of the GAM is illustrated, where the black arrows indicate the direction of the model's forward propagation and the yellow arrows represent the direction of the gradient's backward propagation. By computing the product of the weight matrix of the activation function and the gradient, followed by a weighted summation and processing through the ReLU activation function, we can obtain the SSAM with the same shape as the original image.

that the activation weights obtained by the GAM module are as similar as possible to those during the actual pretraining process. The loss function is defined as follows:

$$L_{\text{SSR}} = \|I_r - \text{concat}\,(I_{m1}, I_{m2})\|_2^2 \tag{2}$$

where I_{m1} and I_{m2} denote multimodal data, and I_r represents the reconstructed image.

As illustrated in Fig. 3, within the GAM framework, black arrows denote the direction of forward model propagation, while yellow arrows represent the backward propagation of gradients. We apply a zero-masking operation to the input multimodal data—specifically, a matrix with all elements set to zero, designed to ensure unobstructed computation of activation weights for every spatial pixel and spectral band. This approach mitigates potential loss of weight information caused by data-obscuring masking strategies.

GAM leverages gradient information from the final convolutional layer to assign spatially resolved importance scores

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to individual neurons relevant to the reconstruction goal. This choice is driven by the unique properties of the final reconstruction convolutional layers: after a series of convolutional and nonlinear transformations, these layers balance abstract semantic representations and fine-grained spatial details. They preserve rich spatial context through convolutional operations while enhancing high-level semantic encoding, making them optimal for extracting fused semantic-spatial features. In contrast, shallow layers primarily capture low-level features (e.g., edges) irrelevant to semantic reconstruction.

The gradient-guided masking logic of GAM is derived from Grad-CAM [44], which extends its gradient-based idea from classification tasks to reconstruction tasks. Goodfellow et al. [45] demonstrate that the gradient of SSR Loss can quantify the contribution of features to reconstruction, while Saxe et al. [46] confirm that models prioritize the optimization of high-gradient "task bottleneck features." Thus, GAM captures hard-to-reconstruct regions through the reconstruction proxy task.

As illustrated in Fig. 3, generating the weight activation map with spatial dimensions $W \times H$ and channel k based on L_{SSR} involves directly calculating gradients of L_{SSR} (prior to the softmax operation) with respect to the feature map activations A_{ij}^{k} at each spatial location (i, j) and channel k, expressed as follows:

$$\alpha_{ij,k}^{L_{\text{SSR}}} = \underbrace{\frac{\partial y^{L_{\text{SSR}}}}{\partial A_{ij}^{k}}}_{\text{gradients via backprop}} \tag{3}$$

where $\alpha_{ij,k}^{L_{\text{SSR}}}$ represents the importance weight of the neuron at (i, j) in channel k. $y^{L_{\text{SSR}}}$ is the scalar output of the SSR Loss function L_{SSR} and A_{ij}^{k} represents the activation value of the feature map at spatial coordinate (i, j) and channel k in the final reconstruction convolutional layer. Unlike traditional methods [44], no global average pooling is applied to the gradients from the final convolutional layer; instead, the spatial coordinates (i, j) are explicitly retained to preserve fine-grained positional information, ensuring the gradients directly reflect the contribution of each spatial–spectral element to the reconstruction loss, enabling precise localization of critical regions in both spatial and channel dimensions.

When calculating $\alpha_{ij,k}^{L_{\text{SSR}}}$ during the backpropagation of gradients with respect to the activation values at each spatial location (i, j) and channel k, the precise calculation process entails a series of matrix multiplications. Specifically, it involves multiplying the weight matrices by the gradients of the activation functions in sequence, continuing until the gradients reach the final convolutional layer. Since no global-average pooling is used, gradients are computed and stored for each individual spatial position. Therefore, this weight $\alpha_{ij,k}^{L_{\text{SSR}}}$ can be regarded as a local linear approximation of the deep network components that are downstream from the activation A_{ij}^{k} at position (i, j) in channel k. It captures the "significance" of the feature at the specific spatial location (i, j) and channel k for the target loss L_{SSR}. This mechanism directly contributes to enhancing the model's pretraining efficiency by minimizing L_{SSR}.

We perform a weighted combination of forward activation maps at each spatial position (i, j), and follow it by a ReLU to obtain H_{ssam}:

$$H_{\text{ssam}} = \text{ReLU}\left(\sum_{k} \alpha_{ij,k}^{L_{\text{SSR}}} A_{ij}^{k}\right). \tag{4}$$

Notably, this process generates a fine-grained heatmap with identical dimensions to the convolutional feature maps, precisely matching the spatial size of the reconstructed image I_r. For each position (i, j), we pass the linear combination of feature maps through a ReLU activation to produce the SSAM. This step emphasizes regions at specific spatial locations (i, j) that show strong responses to the SSR Loss, as our objective is to optimize $y^{L_{\text{SSR}}}$ by amplifying the weights of these high-impact areas.

To generate the spatial–spectral mask matrices, we first apply global average pooling along both the channel and spatial dimensions to condense multidimensional activation responses. We then identify and mask the regions and spectral bands associated with the top importance weights according to a predefined masking ratio. This strategy ensures that the model is compelled to focus on learning from these critical spatial–spectral regions during training, enhancing its capacity to prioritize informative features for effective reconstruction

$$\begin{aligned} S_{\text{spa}} &= S\left(\text{mean}\left(H_{\text{ssam}}, \text{axis} = 1\right), r_1\right) \\ R_{\text{spa}} &= R\left(\text{mean}\left(H_{\text{ssam}}, \text{axis} = 1\right), 1 - r_1, r_2\right) \\ S_{\text{spe}} &= S\left(\text{mean}\left(H_{\text{ssam}}, \text{axis} = [2, 3]\right), r_1\right) \\ R_{\text{spe}} &= R\left(\text{mean}\left(H_{\text{ssam}}, \text{axis} = [2, 3]\right), 1 - r_1, r_2\right). \end{aligned} \tag{5}$$

Let S be a sampling function that operates on the weights derived from the masking ratio and the localization heatmap H_{ssam}. The function R represents the process of randomly selecting samples from the remaining ones after S has completed its sampling. And axis = 1 corresponds to the spatial height dimension, axis = [2, 3] corresponds to spatial width and height dimensions.

After aligning the mask with multimodal inputs via channel broadcasting, the MIM model prioritizes masking high-SSAM regions and retains random coverage of low-SSAM regions, enhancing its ability to learn robust representations. The combined mask can be expressed mathematically as follows:

$$\begin{aligned} M_{\text{spa}} &= S_{\text{spa}} + R_{\text{spa}} \\ M_{\text{spe}} &= S_{\text{spe}} + R_{\text{spe}}. \end{aligned} \tag{6}$$

To avoid visual interference from random pixels, we visualize the SSAM in the spatial dimension, as shown in Fig. 4 (with $r_1 = 30\%$ used). After generating the mask matrices for both spatial and spectral dimensions, we concatenate the two-modal data along the channel dimension to serve as the input for the encoder. The final input accepted by the model is formulated as follows:

$$\begin{aligned} E_{\text{spa}} &= g_{\text{spa}}\left(\text{pad}\left(\text{concat}\left(I_{m1}, I_{m2}\right) * M_{\text{spa}}\right) + \text{PE}\right) \\ E_{\text{spe}} &= g_{\text{spa}}\left(\text{pad}\left(\text{concat}\left(I_{m1}, I_{m2}\right) * M_{\text{spe}}\right) + \text{PE}\right) \end{aligned} \tag{7}$$

where g_{spa} and g_{spe} represent the spatial and spectral embedding layers, respectively. E_{spa} and E_{spe} represent the embedding vectors in the spatial and spectral encoders, respectively.

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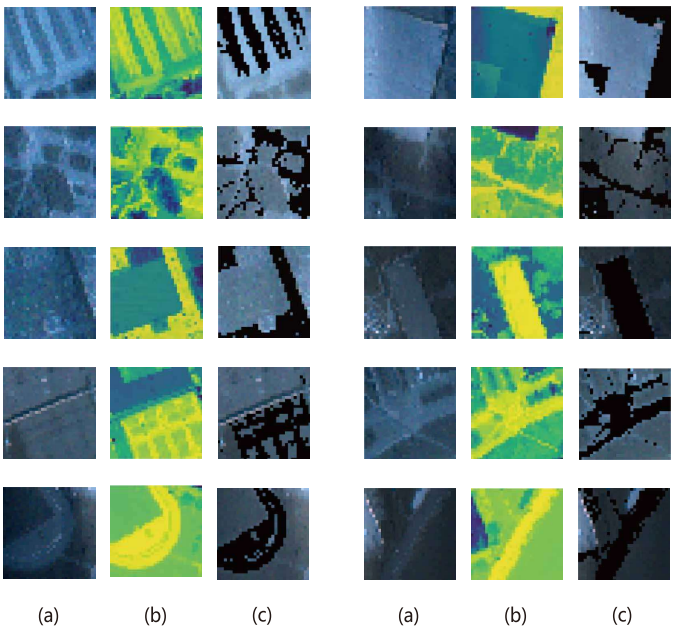

Fig. 4. Visualization of the SSAM in the spatial dimension and the masked images. (a) Original image. (b) SSAM in the spatial dimension. (c) Masked images in the spatial dimension.

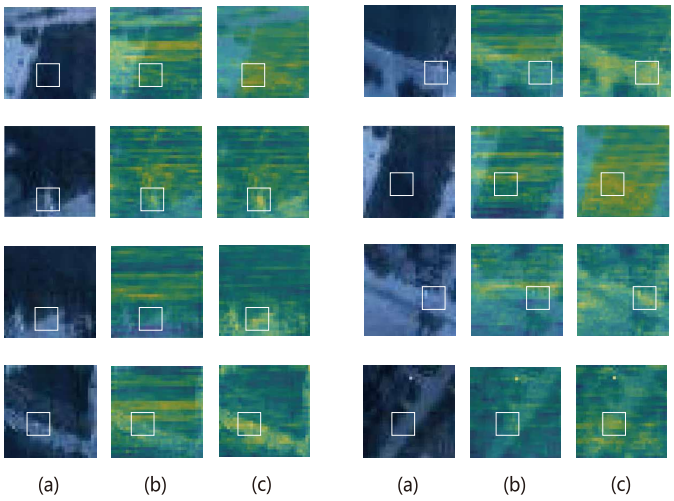

Fig. 5. Visualization of the attention map adjustment results by the ASA mechanism. (a) Original image. (b) Conventional attention heatmap. (c) Attention heatmap refined by the ASA mechanism.

C. ASA Mechanism

In this section, we will provide a detailed introduction to the ASA mechanism. To ensure clarity in the formulation, we will subsequently outline the specific workflow of the spatial branch, which parallels that of the spectral branch. More precisely, we subject the SSAM to the same embedding procedure as applied to the multimodal input data, ultimately yielding H_{ssam}, wherein the masked positions align with those specified in (8).

In the design of attention mechanisms, existing methods still have certain limitations: spatial attention often neglects spectral discriminability, channel attention tends to lose structural details, and hybrid mechanisms lack dynamic spatial–spectral interaction, all of which affect the accurate modeling of complex semantic correlations. Although traditional attention mechanisms are dynamically updated during training, they still struggle to focus on highly discriminative regions in remote sensing scenes with blurred foreground–background boundaries and nonuniform semantic distributions. Notably, GAM has identified key regions (high-reconstruction-loss areas, i.e., "what to learn") via SSAM, while ASA addresses "how to learn" by integrating SSAM's semantic information to dynamically adjust attention weights, thereby enhancing targeted focus on key regions and improving the relevance of feature learning

$$E_{\text{ssam}} = f_{\text{spa}}\left(\text{pad}\left(H_{\text{ssam}}\right) * M_{\text{spa}}\right) + \text{PE} \tag{8}$$

where E_{ssam} represents the embedding vectors in the spatial encoder and PE represents positional encoding. Here, we have specifically trained additional embedding layers f_{spa} for spatial dimension to mitigate the inconsistency in sample space between the pretraining and fine-tuning stages that arises from the introduction of SSAM data.

Subsequently, both E_{spa} and E_{ssam} are mapped to queries (Q), keys (K), and values (V) via linear projections

$$\begin{aligned} Q^{\text{spa}} &= W_Q^{\text{spa}} E_{\text{spa}}, \quad Q^{\text{ssam}} = W_Q^{\text{ssam}} E_{\text{ssam}} \\ K^{\text{spa}} &= W_K^{\text{spa}} E_{\text{spa}}, \quad K^{\text{ssam}} = W_K^{\text{ssam}} E_{\text{ssam}} \\ V^{\text{spa}} &= W_V^{\text{spa}} E_{\text{spa}}, \quad V^{\text{ssam}} = W_V^{\text{ssam}} E_{\text{ssam}} \end{aligned} \tag{9}$$

where W_Q^{spa}, W_K^{spa}, and W_V^{spa} denote the learnable linear projection weights for the spatial branch. Then, on top of the original attention mechanism, we additionally utilize Q^{spa} to compute attention with K^{ssam} and V^{ssam}, enabling the query from the original image to capture weight information from different spatial locations.

ASA integrates SSAM's semantic weights into key–value pairs. In the encoder, it dynamically adjusts the attention weights of regions semantically associated with the high-reconstruction-loss "hard" regions identified by GAM, enabling the encoder to indirectly learn features of "hard" regions through contextual correlations between non "hard" and "hard" regions. The decoder, without masking SSAM, also integrates its semantic information to optimize key–value interactions, thus focusing more precisely on "hard" regions for reconstruction. For instance, in grass-road transition zones, the increased weights of non "hard" healthy grass patches not only assist the encoder in capturing edge-related features but also help the decoder reconstruct adjacent "hard" mixed boundary patches more accurately.

In Fig. 5 (showing decoder attention heatmaps), the ASA-improved heatmap (c) exhibits enhanced focus on critical boundary regions compared to the traditional one (b). This reflects ASA's core function: leveraging SSAM's semantic distribution to prioritize GAM-identified key regions, strengthening feature responses to challenging areas like boundaries, and enabling fine-grained spectral–spatial correlation modeling. Additionally, ASA reduces fluctuations in reconstruction loss during training, leading to a more stable training process and a certain improvement in the reconstruction accuracy of

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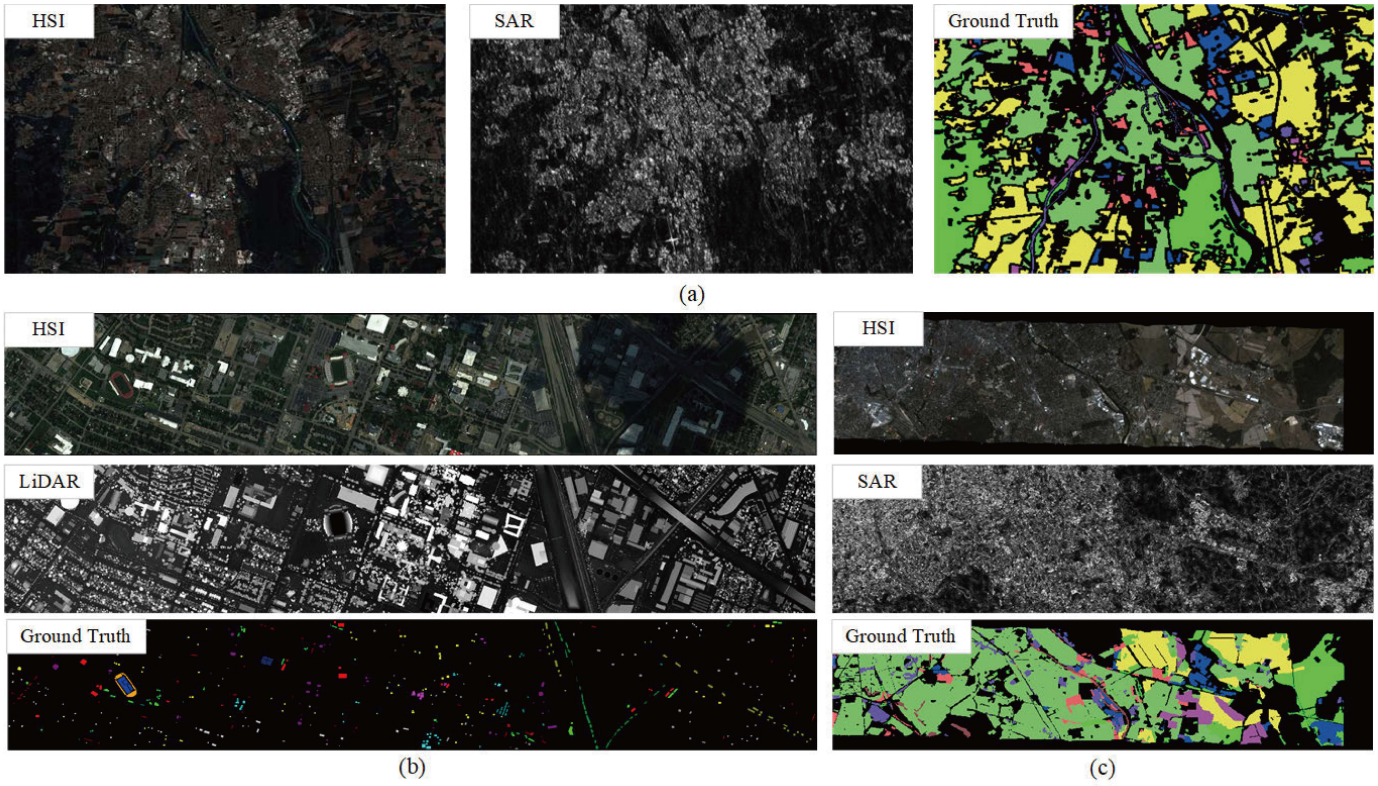

Fig. 6. Visualizationof the MRSI datasets. (a) Augsburg dataset. (b) Houston2013 dataset. (c) Berlin dataset.

"hard" regions

$$\text{Attention}_{\text{ssam}} = \text{softmax}\left(\frac{Q^{\text{spa}}\left(K^{\text{ssam}}\right)^{\top}}{\sqrt{d_k}}\right)V^{\text{ssam}}$$
$$\text{Attention}_{\text{spa}} = \text{softmax}\left(\frac{Q^{\text{spa}}\left(K^{\text{spa}}\right)^{\top}}{\sqrt{d_k}}\right)V^{\text{spa}}. \quad (10)$$

It is worth noting that there is a slight difference between the decoder and encoder: the decoder does not require masking operations on H_{ssam}.

D. Fusion and Reconstruction

To ensure that the fused features retain both spatial–spectral integrity and semantic discriminability, we first project the spatial feature F_{spa} and spectral feature F_{spe} into a shared embedding space through lightweight convolutional layers f_{conv}. These layers facilitate effective cross-modal feature alignment by reducing the dimensional discrepancy between spatial and spectral features and encoding contextual correlations between features.

The projected features engage in fine-grained interaction via elementwise multiplicative operation, generating a correlation map where each entry quantifies the relevance between spatial location (i, j) and spectral band k. As defined in (11), fusion weights α are dynamically derived through a sigmoid activation $\sigma(\cdot)$, which normalizes the correlation map into a probabilistic distribution over $[0, 1]$

$$\alpha = \sigma\left(f_{\text{conv}}\left(F_{\text{spa}}\right) \odot f_{\text{conv}}\left(F_{\text{spe}}\right)\right)$$
$$F_{\text{fused}} = \alpha \odot F_{\text{spa}} + (1-\alpha) \odot F_{\text{spe}} \quad (11)$$

where F_{spa} and F_{spe} represent the feature maps extracted by the spatialwise encoder and spectralwise encoder, f_{conv} denotes a lightweight convolutional block consisting of a 3×3 convolution, batch normalization, and ReLU activation, which projects F_{spa} and F_{spe} into a shared embedding space. This adaptive scheme enables the model to learn coordinate-specific modality importance: $\alpha_{ij,k}$ close to 1 emphasizes spatial features at (i, j, k), while values near 0 prioritize spectral characteristics. By harmonizing spatial structures and spectral signatures through dynamic weighting, the fused representation balances contextual completeness with discriminative power for downstream tasks.

The final reconstruction is obtained by weighted fusion

$$L = \|F_{\text{fused}} - \text{concat}\left(I_{m1}, I_{m2}\right)\|_2^2 \quad (12)$$

where I_{m1} and I_{m2} represent the input data from two modalities, and the loss function employs MSE loss.

IV. Experimental

A. Dataset Description

1) Houston2013 HSI-LiDAR Dataset: Captured using the ITRES CASI-1500 sensor, this dataset focuses on the University of Houston campus and its surrounding areas. It has a total of 144 spectral bands, with wavelengths spanning from 0.38 to 1.05 micrometers. The size of the HSI and LiDAR data is 349×1905 pixels, achieving a spatial resolution of 2.5 m, as illustrated in Fig. 6(b). This dataset is effectively used to categorize the terrain into 15 unique land cover classes.

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TABLE I

LAND COVER CLASSES, NUMBER OF LABELED TRAINING AND TESTING SAMPLES, AND COLOR IN CLASSIFICATION MAP FOR THREE DATASETS, INCLUDING THE HOUSTON2013, AUGSBURG, AND BERLIN DATASETS

Class No.	Houston2013				Augsburg				Berlin			
	Class name	Training	Testing	Color	Class name	Training	Testing	Color	Class name	Training	Testing	Color
1	Healthy Grass	10	1241		Forest	10	13497		Forest	10	54944	
2	Stressed Grass	10	1244		Residential Area	10	30319		Residential Area	10	268632	
3	Synthetic Grass	10	687		Industrial Area	10	3841		Industrial Area	10	19556	
4	Tree	10	1234		Low Plants	10	26847		Low Plants	10	59272	
5	Soil	10	1232		Allotment	10	565		Soil	10	17416	
6	Water	10	315		Commercial Area	10	1635		Allotment	10	13295	
7	Residential	10	1258		Water	10	1520		Commercial Area	10	24814	
8	Commercial	10	1234						Water	10	6662	
9	Road	10	1242									
10	Highway	10	1217									
11	Railway	10	1225									
12	Parking Lot1	10	1223									
13	Parking Lot2	10	459									
14	Tennis Court	10	418									
15	Running Track	10	650									
Total		150	14879			70	78224			80	464591	

2) Augsburg HSI-SAR Dataset [47]: Originating from Augsburg, Germany, this dataset was collected using the HySpex sensor for HSI and the Sentinel-1 sensor for dual-polarimetric synthetic aperture radar (PolSAR). It has spatial dimensions of 332 × 485 pixels, with a resolution of 30 m, as illustrated in Fig. 6(a). The HSI component includes 180 spectral bands, spanning 0.4 to 2.5 micrometers. On the other hand, the PolSAR data includes four distinct features. Overall, this dataset is used to classify the area into seven land cover classes.

3) Berlin HSI-SAR Dataset [47]: This dataset captures the geomorphological features of both urban and rural areas around Berlin. It integrates synthesized EnMAP and HyMap HSI datasets for HSI data and uses Sentinel-1 dual-polarization (VV-VH) single-look complex (SLC) product from the European Space Agency (ESA) for SAR information. The HSI dataset consists of 797 × 220 pixels with 244 spectral bands, spanning 0.4–2.5 micrometers, which provides a spatial resolution of 30 m, as illustrated in Fig. 6(c). On the contrary, the SAR dataset has an image resolution of 1723 × 476 pixels, at a finer spatial resolution of 13.89 m, and is used to categorize eight distinct land cover types. The HSI data are aligned to the SAR image dimensions by means of nearest-neighbor interpolation.

B. Experimental Setting

1) Evaluation Metrics: Throughout the experiments, four quantitative assessment indices are utilized to gauge the MRSI classification performance. These specifically encompass per-category accuracy, overall accuracy (OA), average accuracy (AA), and the Kappa coefficient.

2) Implementation Details: All experiments were conducted using the PyTorch framework on a computer equipped with an NVIDIA 4090 GPU featuring 24 GB of memory. The Adam optimizer was used to optimize the proposed model. During the self-supervised mask reconstruction pretraining phase, the batch size was set to 32, the learning rate was maintained at 0.001, and a total of 300 epochs were performed; 10% of the unlabeled samples from the dataset were selected for the pretraining stage. In the fine-tuning phase, both the batch size and learning rate remained unchanged, with an additional 300 epochs conducted. For each dataset, ten labeled samples per class were randomly selected as training samples for the fine-tuning phase. Ten training runs were performed using random seeds from 0 to 9, and the results were aggregated as the final outcome. Detailed information on the training and testing samples is provided in Table I. Additionally, to mitigate noise interference in real-world datasets (e.g., strip noise and cloud-shrouded interference), we first applied principal component analysis (PCA) to the original HSI data, reducing the spectral dimensions to 32. This step not only helps alleviate noise-related issues but also contributes to reducing computational complexity. Meanwhile, the patch size of magnetic resonance spectroscopic imaging (MRSI) was set to 11.

C. Comparison to State-of-the-Art Works

To evaluate the effectiveness of our proposed method from multiple perspectives, we established a comparative experimental framework encompassing three dimensions: single-modality self-supervised learning, multimodal supervised learning, and multimodal self-supervised learning.

In the realm of single-modality self-supervised contrastive models, we focus on two notable approaches. SimCLR [48], utilizing a ResNet-50 backbone, extracts visual features within a contrastive learning framework by enhancing semantic similarity across augmented views of the same image, thereby improving representation robustness. GraphGST [49] introduces a novel graph-based structure perception mechanism, employing pixel-level graph embeddings to replace traditional patch-based contrastive strategies, which enhances the differentiation of heterogeneous pixels.

Within the domain of multimodal supervised learning contrastive models, MIFNet [50] and NCGLF2 [51] stand out. MIFNet integrates an adaptive filter with a window attention mechanism to fuse frequency and feature information, using a cross-scale feature correlation matrix to minimize spatial information loss. NCGLF2 combines the strengths of CNN and Transformer architectures, adopting a hierarchical global–local feature processing strategy to achieve efficient fusion of multimodal remote sensing data.

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TABLE II

OA, AA, κ, AND THE ACCURACY PER CLASS FOR DIFFERENT METHODS ON AUGSBURG DATASET

Class	self-supervised&single-modality		supervised&multimodal		self-supervised&multimodal					
	SimCLR [48]	GraphGST [49]	MIFNet [50]	NCGLF2 [51]	CMCL [52]	SS-MAE [39]	LDS2AE [53]	UACL [54]	DAE-GSP [55]	Proposed
1	83.74±6.71	79.76±7.53	82.48±9.40	**92.29±3.47**	84.65±10.29	89.23±5.24	90.16±4.49	88.44±5.46	90.46±4.67	90.93±2.03
2	78.10±5.11	92.44±2.22	89.15±7.79	94.65±3.75	93.44±4.03	93.85±1.63	**97.28±1.87**	91.17±3.15	92.25±2.97	90.04±1.70
3	27.85±5.42	59.08±7.58	52.29±10.40	**64.52±14.55**	62.47±5.03	59.72±5.85	47.22±9.45	54.49±5.76	63.66±8.36	59.33±8.76
4	82.21±8.86	90.44±5.07	91.81±7.00	96.88±1.92	**99.24±1.44**	93.21±3.36	95.08±2.99	91.35±3.70	93.73±3.33	89.40±1.32
5	6.12±3.20	8.66±3.62	4.79±1.71	9.38±3.88	6.49±1.64	12.16±3.67	10.39±4.35	13.26±5.47	12.00±2.56	**25.35±5.50**
6	5.38±2.16	19.19±4.47	14.53±6.04	17.22±6.31	30.65±7.05	26.47±5.68	20.95±5.34	19.91±3.52	27.13±8.57	**38.83±2.66**
7	4.82±1.88	9.36±2.79	22.83±12.73	34.49±21.23	40.42±7.22	32.09±11.83	33.47±23.15	38.98±19.41	**44.32±11.28**	44.23±7.66
OA	61.96±2.13	78.71±3.05	66.08±6.61	77.37±5.19	79.16±1.90	81.42±2.73	78.33±2.50	80.88±2.65	82.64±1.78	**84.69±1.23**
AA	43.56±1.86	56.93±3.48	63.25±2.04	68.88±4.64	59.68±4.82	**72.03±1.05**	70.69±2.67	64.33±3.06	72.01±1.64	69.81±2.81
$\kappa\times100$	49.15±2.57	70.55±4.09	56.45±7.64	69.90±6.21	71.84±2.67	74.61±3.36	71.16±3.36	73.54±3.35	76.04±2.26	**79.28±3.16**

TABLE III

OA, AA, κ, AND THE ACCURACY PER CLASS FOR DIFFERENT METHODS ON HOUSTON2013 DATASET

Class	self-supervised&single-modality		supervised&multimodal		self-supervised&multimodal					
	SimCLR [48]	GraphGST [49]	MIFNet [50]	NCGLF2 [51]	CMCL [52]	SS-MAE [39]	LDS2AE [53]	UACL [54]	DAE-GSP [55]	Proposed
1	**94.45±1.46**	92.56±4.30	91.18±8.01	85.25±8.71	90.66±4.04	91.98±6.77	90.27±6.43	93.69±4.67	93.26±4.53	89.50±1.71
2	**93.97±2.81**	92.21±2.59	92.23±6.66	88.11±8.02	86.39±6.39	91.99±5.78	87.69±5.81	92.37±5.41	93.17±5.30	90.20±1.97
3	**99.54±1.02**	95.12±4.10	85.58±10.89	99.33±0.91	99.53±0.92	95.13±5.58	94.74±0.60	97.88±5.73	96.56±6.27	98.98±1.83
4	97.46±0.90	92.00±4.65	89.25±5.05	84.03±6.95	89.55±5.11	94.08±3.32	90.00±1.00	**98.44±1.00**	96.35±2.73	95.07±0.81
5	97.93±0.89	93.91±5.65	97.27±6.41	95.29±6.58	88.90±6.28	**98.63±1.96**	97.39±1.54	97.32±2.37	97.79±2.25	**98.63±2.28**
6	10.90±2.86	80.03±15.40	62.67±15.78	95.54±5.45	95.19±2.54	94.78±15.16	84.45±12.73	**99.38±0.64**	99.14±1.18	94.30±2.59
7	76.49±2.39	78.77±8.45	83.51±6.16	83.55±6.35	86.73±5.61	91.52±3.04	92.68±2.26	91.84±4.56	91.12±2.21	**93.18±1.72**
8	81.36±3.38	78.53±4.96	86.03±10.21	**90.18±3.79**	87.00±5.74	82.88±8.37	89.19±7.85	84.76±6.10	83.90±7.24	86.73±1.22
9	78.37±4.06	80.98±4.01	74.19±7.65	73.49±14.28	76.63±5.22	84.16±9.82	84.46±3.32	81.15±11.11	84.63±8.57	**86.62±3.83**
10	**89.84±7.51**	72.44±7.35	68.82±7.00	72.61±13.83	77.44±3.88	72.38±7.95	84.80±6.09	75.92±6.43	79.02±4.51	83.37±3.78
11	82.71±1.59	86.66±4.79	76.09±6.60	78.12±7.78	84.96±4.10	86.27±6.39	90.46±3.86	85.51±4.78	89.65±4.37	**94.05±1.56**
12	86.60±4.05	80.17±6.37	74.40±6.12	72.25±10.10	73.89±4.69	81.47±7.20	82.34±3.62	77.96±8.55	82.77±6.72	**86.70±3.12**
13	57.09±3.43	66.61±13.33	63.32±9.76	72.59±13.03	89.97±6.89	94.21±3.44	74.09±9.77	**94.91±4.52**	93.77±8.89	85.56±4.29
14	87.41±3.22	89.14±9.50	62.51±11.61	98.68±2.90	96.94±1.44	95.57±6.56	90.74±6.32	97.19±2.78	97.48±3.02	**99.21±1.31**
15	93.92±9.64	86.08±7.66	98.19±10.06	94.93±1.77	97.81±0.86	96.61±2.19	94.74±1.87	95.38±3.04	97.57±2.05	**99.05±0.76**
OA	78.31±1.17	84.89±1.04	80.65±1.63	82.80±2.14	85.92±1.57	88.02±1.48	88.68±1.54	88.84±1.23	90.15±1.37	**91.24±1.13**
AA	81.81±1.20	86.36±1.51	82.45±1.57	85.37±1.91	88.17±1.17	89.58±1.27	90.05±1.38	89.95±1.11	91.33±1.15	**92.41±0.89**
$\kappa\times100$	76.67±1.23	83.61±1.13	79.13±1.75	81.41±2.31	84.77±1.70	87.00±1.60	87.78±1.66	87.93±1.34	89.35±1.47	**90.73±1.10**

TABLE IV

OA, AA, κ, AND THE ACCURACY PER CLASS FOR DIFFERENT METHODS ON BERLIN DATASET

Class	self-supervised&single-modality		supervised&multimodal		self-supervised&multimodal					
	SimCLR [48]	GraphGST [49]	MIFNet [50]	NCGLF2 [51]	CMCL [52]	SS-MAE [39]	LDS2AE [53]	UACL [54]	DAE-GSP [55]	Proposed
1	60.52±10.95	81.61±5.02	79.31±8.54	71.58±8.82	77.28±5.82	**84.14±6.44**	76.20±9.08	81.93±9.88	83.77±6.00	79.41±4.04
2	84.39±1.98	**93.65±0.91**	91.64±1.83	92.05±1.39	87.55±1.34	91.32±2.52	89.02±2.93	90.48±2.07	93.05±2.11	82.94±3.49
3	45.51±11.48	27.97±4.26	29.59±8.67	**50.10±10.95**	44.04±9.87	36.73±9.35	28.55±11.77	39.89±7.62	40.63±5.94	42.32±5.78
4	62.03±4.10	69.15±7.04	82.18±6.18	81.58±10.62	**84.40±4.00**	64.10±5.67	79.28±4.94	79.65±9.54	70.94±2.14	77.10±1.99
5	43.80±5.33	41.80±5.32	60.02±7.65	61.96±14.39	**89.40±6.48**	62.02±4.62	59.23±17.10	55.86±16.73	60.92±5.05	73.38±6.50
6	13.36±2.62	28.71±11.38	21.06±8.55	19.92±5.29	12.82±1.95	16.79±2.56	14.19±13.71	19.50±6.14	21.19±2.24	**33.42±4.77**
7	22.12±2.76	23.08±6.50	21.05±4.83	29.42±6.00	24.07±2.60	21.61±7.13	21.93±10.20	25.89±8.16	21.20±6.88	**41.10±5.40**
8	7.47±1.45	**65.26±18.05**	19.01±6.16	16.81±7.36	50.14±15.50	62.27±6.55	19.25±10.03	48.71±24.26	60.74±6.13	58.07±3.89
OA	55.65±3.22	69.05±1.68	61.72±4.83	65.62±3.64	64.95±3.32	66.09±2.47	68.79±3.73	69.55±2.56	71.03±2.64	**73.43±2.78**
AA	47.34±2.19	64.45±2.47	65.17±2.16	66.49±2.39	55.47±2.77	58.42±2.30	61.32±3.86	56.38±4.93	66.70±2.27	**70.83±4.65**
$\kappa\times100$	38.05±2.57	56.09±1.81	48.74±4.74	52.84±3.87	49.61±3.49	50.74±2.51	54.24±4.41	53.83±3.87	58.05±2.47	**61.45±2.64**

In the field of multimodal self-supervised methods, we selected CMCL [52], SS-MAE [39], LDS2AE [53], UACL [54], and DAE-GSP [55] as baseline models. CMCL captures cross-model consistency and complementarity through feature alignment. SS-MAE employs a spatial–spectral joint masked autoencoder to extract correlated features. LDS2AE leverages a local diffusion shared-specific autoencoder with masking to model data distributions. UACL enhances spectral–spatial feature discriminability via mixed contrastive learning and a three-branch representation. DAE-GSP proposes a patch selection mechanism, integrating contrastive learning and MIM to boost feature discriminability.

This structured experimental design enables a comprehensive assessment of our method's performance across supervised, self-supervised, single-modality, and multimodal paradigms.

1) Quantitative Evaluation: For various methods, OA, AA, κ, and the accuracy of each class are shown in Tables II, III, and IV. Each experiment is carried out ten times, and in the experiments, the mean value and standard deviation of each index are presented.

Self-supervised learning methods are capable of learning to extract general features from large amounts of unlabeled data. Consequently, under conditions of limited training data, self-supervised learning models (e.g., GraphGST) exhibit superior convergence and stability compared to supervised learning models (e.g., NCGLF2). For instance, on the Berlin dataset in Table IV, the GraphGST model demonstrates a standard deviation of 1.68%, which is significantly lower than the 3.64% observed for NCGLF2.

When only ten labeled samples are available per class, models that integrate self-supervised learning with multi-

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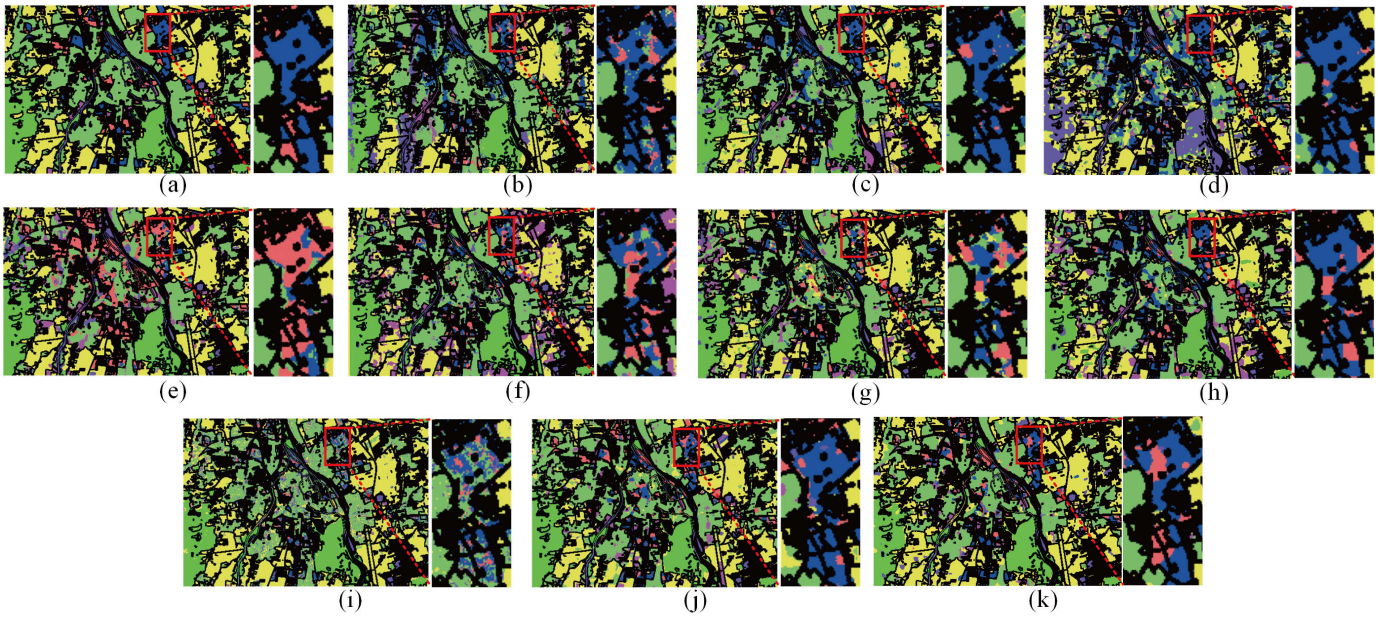

Fig. 7. Classification maps of different methods on the Augsburg dataset. (a) Ground truth. (b) SimCLR (OA = 61.96%). (c) GraphGST (OA = 78.71%). (d) MIFNet (OA = 66.08%). (e) NCGLF2 (OA = 77.37%). (f) CMCL (OA = 79.16%). (g) SS-MAE (OA = 81.42%). (h) LDS2AE (OA = 78.33%). (i) UACL (OA = 80.88%). (j) DAE-GSP (OA = 82.64%). (k) Proposed (OA = 84.69%).

modal approaches typically achieve enhanced performance. For example, on the Houston2013 dataset, although the CMCL method yields a relatively low OA, it surpasses the NCGLF2 method by 3.12%.

Among high-performing multimodal self-supervised methods, quantitative comparisons reveal that the method proposed in this article significantly outperforms other models across key metrics, including OA, AA, and κ. This is due to the GAM module's ability to capture "hard regions"—those with high reconstruction loss, such as spectrally ambiguous boundaries and complex structural areas—through SSAM. This allows the pretraining process to focus on critical information, preventing random masking from overlooking important features and ensuring robust recognition of complex ground features even when sample sizes are small.

On the Augsburg dataset, the proposed model performs comparably to its performance on the Berlin dataset, outperforming existing models significantly in challenging categories (Classes 5 and 6), and achieving the highest OA of 84.69%, a 2.05% improvement over the state of the art. On the Houston2013 dataset, the model excels in nearly half of the categories, maintaining a minimum category accuracy of 83.37% (while other models are below 80%) and leading across all evaluation metrics. On the Berlin dataset, the model outperforms existing methods in difficult categories (for instance, 33.42% $\pm$ 4.77% in Class 6 versus GraphGST's 28.71% $\pm$ 11.38%; 41.10% $\pm$ 5.40% in Class 7 versus NCGLF2's 29.42% $\pm$ 6.00%) and leads in all comprehensive metrics, with an OA 2.4% higher than the best existing model.

2) Statistical Significance Analysis: To verify the statistical reliability of the proposed method's performance improvements, we conducted rigorous tests. Each dataset was trained ten times with random seeds 0–9 to reduce random errors. After verifying data normality via the Shapiro–Wilk test, we used paired t-tests, Wilcoxon signed-rank tests, and Bonferroni correction (=0.0167) to assess significance, with Cohen's d measuring effect size.

The results show the following: on Berlin, OA, AA, and Kappa had $p < 0.0167$ and large effect sizes (1.1287–1.331); on Houston2013, OA and Kappa had $p < 0.0167$ and medium-to-large effects (0.8718–1.0623); on Augsburg, OA and Kappa had $p < 0.0167$ and large effects (1.6087–1.6718). These confirm that the 1.09%, 2.05%, and 2.40% improvements over SOTA methods are statistically significant.

3) Visual Evaluation: The classification maps of the three datasets generated by different methods are shown in Figs. 7–9. The semantic information provided by a large number of unlabeled samples allows self-supervised methods to produce more accurate and realistic classification maps. More importantly, the structural information provided by the generative method has been shown to be effective in distinguishing some similar classes. This is evident in the highlighted area of Fig. 10 when differentiating between an industrial area (blue) and a commercial area (red). Guided by both discriminative and structural information, our method achieves more faithful visual results.

D. Ablation Studies

1) Impact of the GAM: In this section, we investigate the effects of different mask ratios, pretraining epochs, the update frequency of the SSAM, and varying amounts of pretraining data on experimental results, aiming to determine the optimal configuration for the SSAM-driven masking approach and analyze its advantages over random masking.

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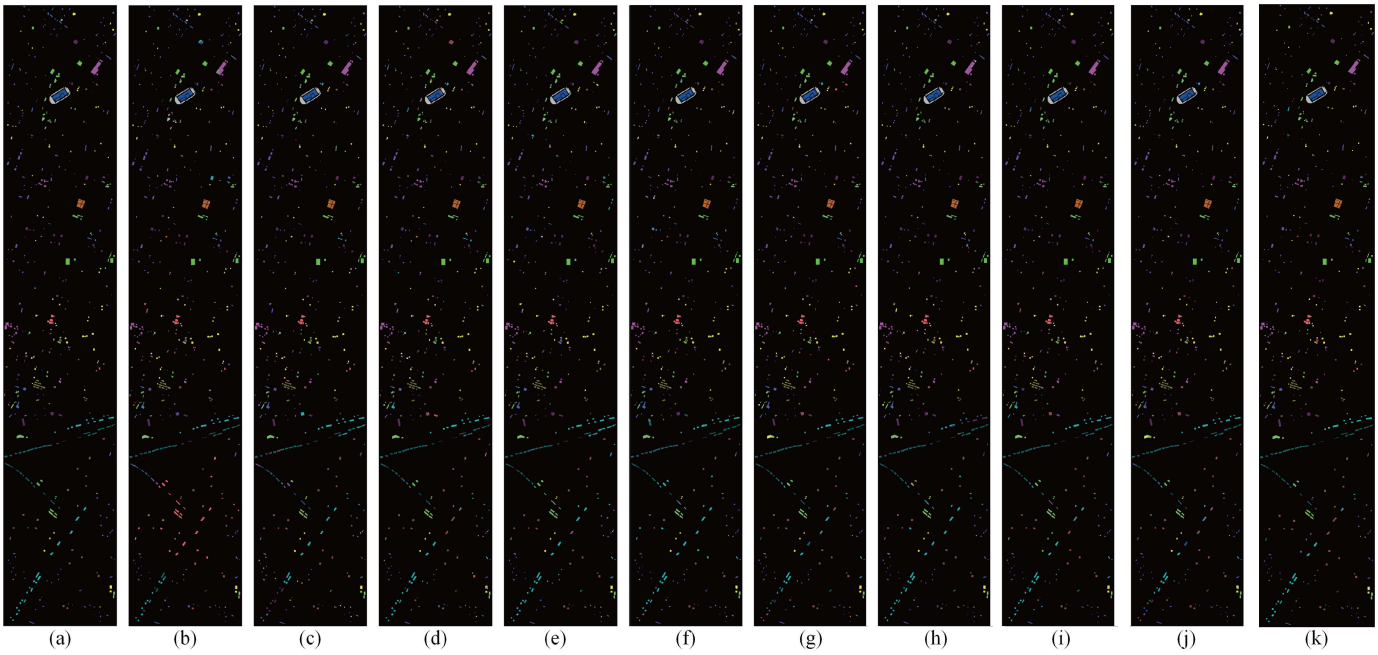

Fig. 8. Classification maps of different methods on the Houston2013 dataset. (a) Ground truth. (b) SimCLR (OA = 78.31%). (c) GraphGST (OA = 84.89%). (d) MIFNet (OA = 80.65%). (e) NCGLF2 (OA = 82.80%). (f) CMCL (OA = 85.92%). (g) SS-MAE (OA = 88.02%). (h) LDS2AE (OA = 88.68%). (i) UACL (OA = 88.84%). (j) DAE-GSP (OA = 90.15%). (k) Proposed (OA = 91.24%).

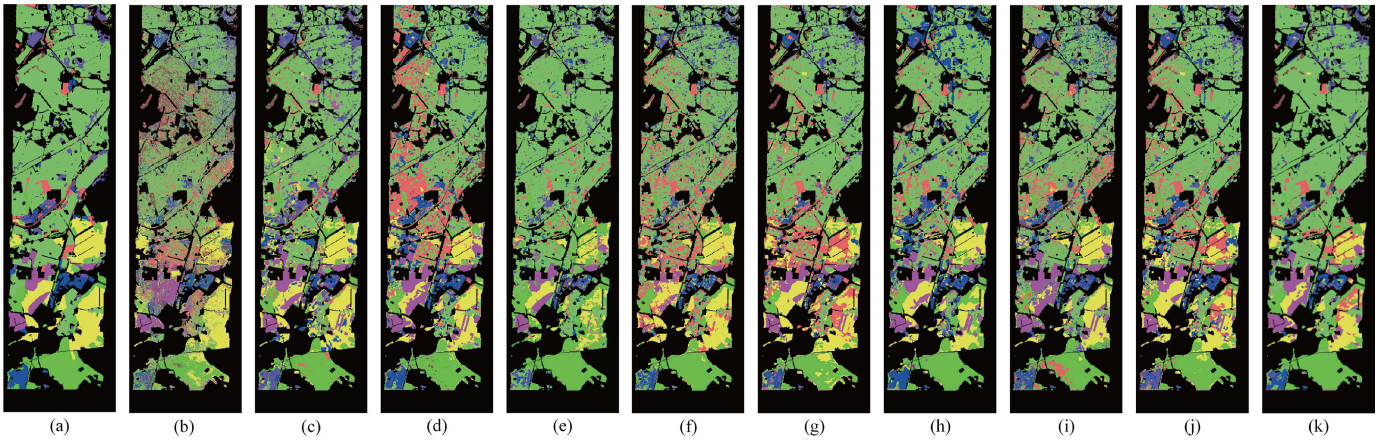

Fig. 9. Classification maps of different methods on the Berlin dataset. (a) Ground truth. (b) SimCLR (OA = 55.65%). (c) GraphGST (OA = 69.05%). (d) MIFNet (OA = 61.72%). (e) NNCNet (OA = 64.95%). (f) CMCL (OA = 66.09%). (g) SS-MAE (OA = 68.79%). (h) LDS2AE (OA = 69.55%). (i) UACL (OA = 69.12%). (j) DAE-GSP (OA = 71.03%). (k) Proposed (OA = 73.43%).

TABLE V

INFLUENCE OF DIVERSE MASKING RATIOS ON OA FOR THREE DATASETS: HOUSTON2013, AUGSBURG, AND BERLIN

Ratio	Berlin	Houston2013	Augsburg
10%	72.72	90.66	84.31
20%	73.21	90.89	84.54
30%	**73.43**	**91.24**	84.46
40%	72.93	91.03	**84.69**
50%	72.39	90.63	84.35

TABLE VI

INFLUENCE OF DIFFERENT RANDOM PERTURBATION MASKING RATIOS ON OA FOR THREE DATASETS: HOUSTON2013, AUGSBURG, AND BERLIN

Ratio	Berlin	Houston2013	Augsburg
1%	72.99	90.67	84.61
3%	73.24	91.04	**84.69**
5%	**73.43**	**91.24**	84.53
7%	72.75	91.13	84.03
9%	72.90	90.69	84.29

a) Different mask ratios: In the experiments conducted on three datasets, to explore the optimal masking ratio as defined in $r_1 + r_2$, where r_1 is the target masking ratio and r_2 is the random perturbation ratio, we adjusted the combined ratio from 10% to 50%. As shown in Table V, the model achieved peak performance with a 30% masking ratio on the

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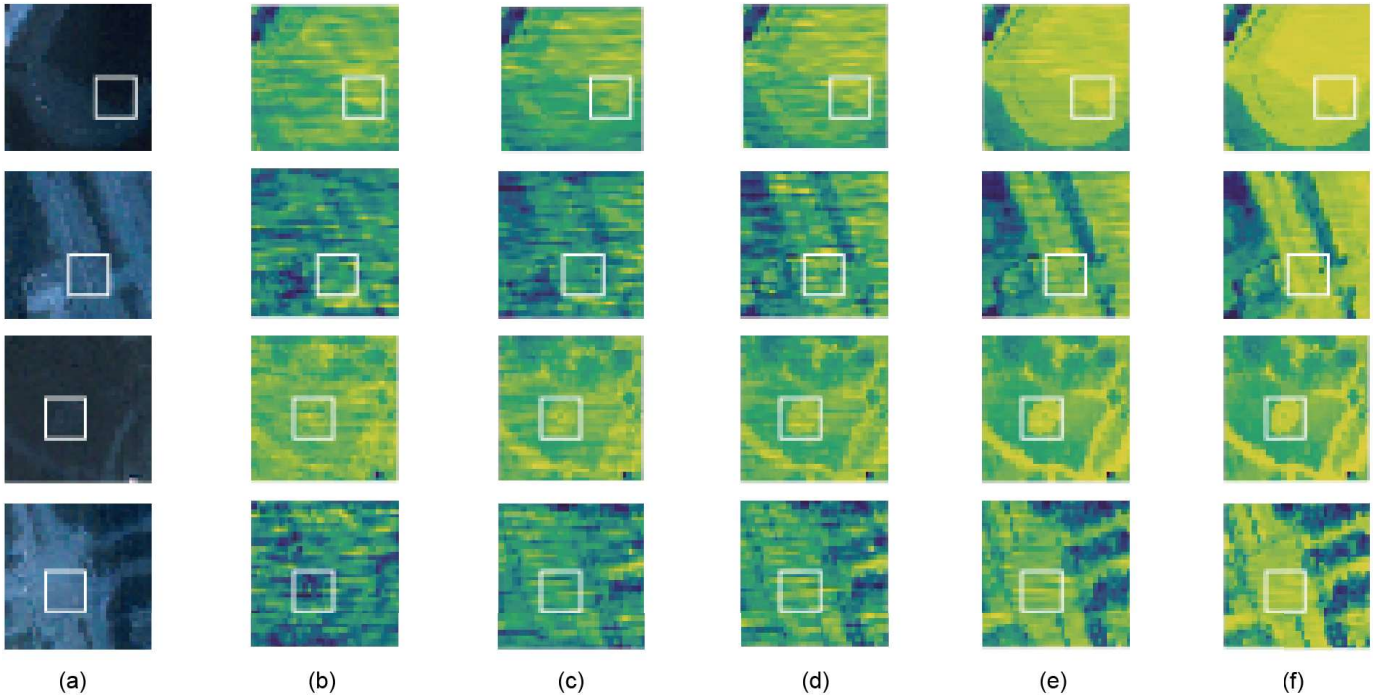

Fig. 10. SSAM visualization in the spatial dimension. For each subplot, visualize the results for (a) original image, (b) 20 epochs, (c) 40 epochs, (d) 60 epochs, (e) 80 epochs, and (f) 100 epochs. The white box highlights the activation weight distribution in the same region across different epochs.

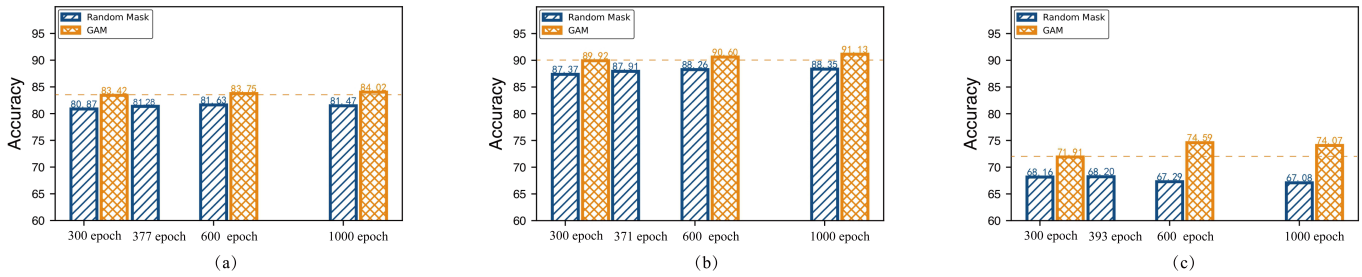

Fig. 11. Classification performance comparison between GAM and random masking under different training epochs. (a) Augsburg. (b) Houston2013. (c) Berlin.

Berlin and Houston2013 datasets, and 40% for the Augsburg dataset. This is because Houston2013 and Berlin have strong spatial–spectral feature correlation, so $r = 30\%$ total masking ratio covers critical regions, while Augsburg's more complex HSI-SAR complementarity requires $r = 40\%$ total masking ratio to avoid missing key regions. The corresponding r_2 optimal ratios for each dataset, which balance contextual information utilization, are detailed in Table VI.

Remote sensing images differ from ordinary traditional visual images in that they have a higher information density. This characteristic makes a 30% masking ratio fall within a reasonable expectation. Analyzing from the in-depth perspective of information utilization, if the masking ratio is set too low, the features learned by the model may be highly redundant, making it difficult to deeply excavate the valuable information hidden in the images. Conversely, if the masking ratio is set too high, the model is likely to lose excessive key information, resulting in difficulty in learning effective feature representations and thus affecting the model's performance. A 30% masking ratio strikes an excellent balance between the two. It can not only ensure that the model learns an adequate amount of key information to meet its requirements for effective features but also skillfully avoid the interference of redundant information, enabling the model to exhibit excellent performance in classification tasks and accurately classify various images.

b) Random mixing strategy: As illustrated in Fig. 10, we present the visualization results of the spatial–spectral activation map (SSAM) from 20 to 100 training epochs. Notably, the SSAM's target region discrimination gradually stabilizes after 100 epochs, causing the masking strategy to consistently focus on target areas while background pixels are significantly underutilized—a phenomenon undesirable for balanced feature learning. To address this, we introduce random perturbations into the masking process to effectively leverage background information. We systematically investigated the effects of 1%, 3%, 5%, 7%, and 9% random masking ratios. As demonstrated in Table VI, a 5% randomness ratio

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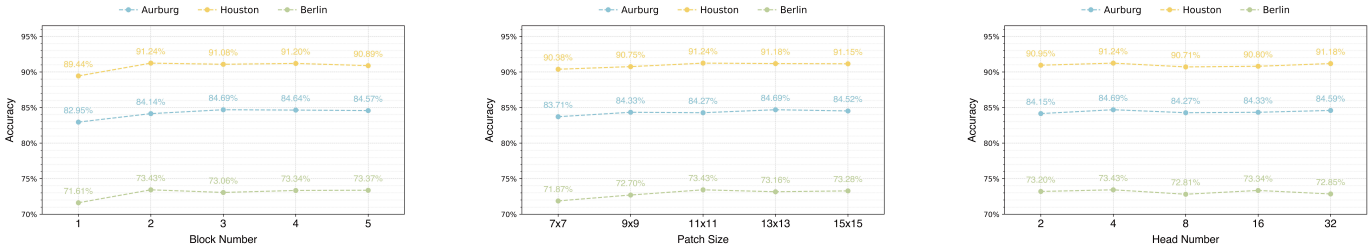

Fig. 12. Impact of model hyperparameters. Impact of (a) ViT patch size. (b) Multihead attention heads. (c) Transformer block number.

TABLE VII

INFLUENCE OF DIVERSE MASKING RATIOS OF RANDOM MASKING ON OA FOR THREE DATASETS: HOUSTON2013, AUGSBURG, AND BERLIN

Ratio	Berlin	Houston2013	Augsburg
10%	67.22	87.03	80.33
30%	**68.16**	**87.37**	80.23
50%	66.63	86.79	80.65
70%	67.08	86.93	**81.06**
90%	67.16	86.45	80.22

TABLE VIII

INFLUENCE OF DIFFERENT EPOCH UPDATE FREQUENCY ON OA FOR THREE DATASETS: HOUSTON2013, AUGSBURG, AND BERLIN

Epoch	Berlin	Houston2013	Augsburg
1	**73.89**	**91.55**	**84.92**
5	73.51	91.47	84.65
10	73.43	91.24	84.69
20	73.03	90.74	83.99
50	72.24	90.03	82.98

yielded optimal performance on the Berlin and Houston2013 datasets, while a 3% ratio was found to be most effective for the Augsburg dataset. This adaptive blending of target-focused and random masking ensures the model can efficiently exploit both discriminative target features and contextual background information, avoiding overconcentration on dominant regions and enhancing overall representation robustness.

c) Performance comparison: As shown in Fig. 10, we visualize the dynamic changes in SSAM weights in the spatial dimension at different training stages (at 20-epoch intervals). In the actual pretraining process, the generation frequency of SSAM is set to update every 10 epochs, a strategy balancing pretraining efficiency and computational cost: on one hand, overly frequent updates (e.g., less than 10 epochs) significantly increase the computational load of pretraining (with the computation amount per updating epoch being 1285.43, much higher than 663.75 for nonupdating epochs); on the other hand, excessively long update intervals (e.g., more than 10 epochs) may cause the model to become stuck in learning features from fixed regions, making it difficult to capture new semantic information. Notably, after 100 training epochs, although the SSAM weight distribution tends to stabilize, the 10-epoch update frequency is maintained to avoid model dependence on fixed masked regions and maintain the dynamism of training objectives.

TABLE IX

EFFECTIVENESS OF ESSENTIAL COMPONENTS FOR THE AUGSBURG DATASET

Model	Params (MB)	FLOPs (MB)
SimCLR	27.803	37.152
GraphGST	1.765	1.769
MIFNet	0.173	11.713
NCGLF2	0.512	76.449
CMCL	0.281	8.413
SS-MAE	4.821	569.09
LDS2AE	0.722	140.178
UACL	0.188	18.605
DAE-GSP	7.60	696.53
GA-MAE	7.96	663.75

d) Update frequency optimization: To balance SSAM's dynamic adaptability—its ability to dynamically adjust to evolving semantic distributions in images as training progresses—and computational efficiency, we tested intervals of 1, 5, 10, 20, and 50 epochs (Table VIII). Results show that OA remains stable for intervals of 10 epochs (73.43%, 91.24%, and 84.69% at 10 epochs for Berlin, Houston2013, and Augsburg, with minimal drops, e.g., 0.46% lower for Berlin at 10 versus 1 epoch), but drops significantly beyond 10 epochs (72.24%, 90.03%, and 82.98% at 50 epochs). This is due to SSAM's semantic capture: overly frequent updates (e.g., 1 epoch) increase computations, while excessively long intervals hinder tracking semantic changes, reducing generalization. Thus, 10 epochs are optimal—balancing efficiency and dynamic focus on high-value regions (e.g., boundaries and spectral confusion areas).

To evaluate the pretraining effectiveness and generalization capability of GAM compared with random masking, we conducted comparative experiments with 300, 600, and 1000 epochs on three datasets: Houston2013, Berlin, and Augsburg. It should be noted that the total masking ratios of the proposed method and random masking are consistent: the masking of the proposed method consists of SSAM-guided target masking (r) and random perturbation masking (r), with their sum being the total masking ratio—30% for the Berlin and Houston2013 datasets, and 40% for the Augsburg dataset—to ensure the fairness of the comparison. As shown in Fig. 11, across different training epochs, the GAM method significantly outperforms the traditional random masking strategy on all datasets.

To verify the source of performance advantages, we conducted ablation experiments with random masking ratios of

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10%, 30%, 50%, 70%, and 90%. Results show that even with optimal random ratios (30% for Berlin, 30% for Houston2013, and 70% for Augsburg), our method still improves by 3.75%, 2.55%, and 2.36%, confirming that advantages are not from ratio differences, as shown in Table VII.

e) Time efficiency comparison: As shown in Fig. 11, under the same experimental settings and identical training time, GAM completed 300 pretraining epochs while random masking required 377, 371, and 393 epochs on Augsburg, Houston2013, and Berlin datasets, respectively. Despite more iterations, random masking's OA was 2.14%, 1.66%, and 3.71% lower. This gap arises from GAM's targeted learning via SSAM (focusing on high-reconstruction-loss regions), enabling faster capture of key features. Random masking, lacking targeting, wastes epochs on redundant regions, even degrading at high epochs (e.g., Berlin). Thus, GAM's extra computational cost is offset by its efficient learning, showing better cost-effectiveness.

Notably, on the Berlin dataset with the highest classification difficulty, the random masking method exhibits performance degradation as the number of epochs increases, while the GAM method still achieves performance improvement by dynamically adjusting the masking strategy, effectively alleviating overfitting issues. Further analysis indicates that due to the high complexity of land cover categories and strong spectral confusion in the Berlin dataset, the performance improvement margin of the GAM is significantly higher than that of other datasets, fully verifying its superiority and robustness in complex remote sensing scenarios.

2) Impact of Model Hyperparameter: We studied the influence of model parameter settings on the performance of remote sensing image classification tasks (including block number, attention head number, and token dimension of the ViT encoder and decoder). To speed up the experiments, we set the hyperparameters of encoder and decoder to the same value. The results can be found in Fig. 12.

ViT Block Number: Performance degrades significantly with 1 block (insufficient depth, hindering learning of discriminative features). With 2 blocks, performance stabilizes (accuracy fluctuations within $< 1\%$), with diminishing marginal benefits from further increasing depth. Optimal block counts vary by dataset: Augsburg requires three blocks (OA 84.69%), while Berlin and Houston2013 each need two blocks (OA 73.43% and 91.24%, respectively), indicating that dataset-specific optimization of block numbers is necessary to boost performance.

Patch Size: Notable performance differences exist across sizes (linked to receptive fields). A 7×7 patch performs poorly on Berlin (small patches contain fewer land cover classes, weakening GAM effectiveness). Optimal sizes: 13 for Augsburg, and 11 for Houston2013 and Berlin, requiring a balance of local details and global context to match land cover distributions and masking strategy needs.

Number of Attention Heads: The model shows low sensitivity to this parameter (accuracy fluctuations within $< 1\%$), with optimal performance at four heads across all three datasets. This indicates that a moderate number of heads suffices for effective feature interactions in multihead attention, with excessive heads yielding no significant gains.

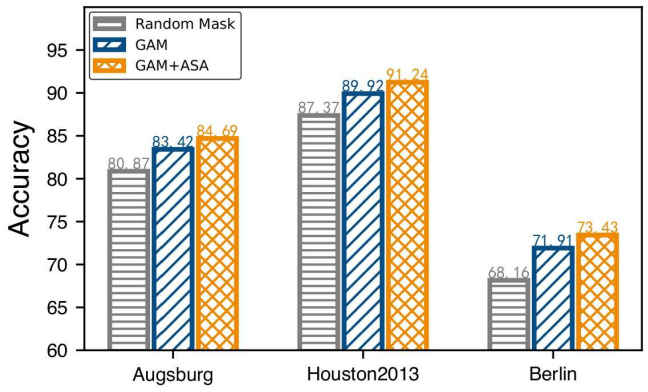

Fig. 13. Effectiveness of core components (GAM and ASA) in datasets Augsburg, Houston2013, and Berlin.

3) Effectiveness of Essential Components: In the pretraining stage, we propose a GAM and design an ASA based on the weight distribution of the spatial–spectral activation of the proposed method is evaluated on three datasets: Augsburg, Houston2013, and Berlin. As shown in Fig. 13, quantitative results show that compared with the random masking strategy, using GAM alone can bring significant performance improvements: on the Augsburg, Houston2013, and Berlin datasets, the OA increases from 80.87 to 83.42 (+2.55%), from 87.37 to 89.92 (+2.55%), and from 68.16 to 71.91 (+3.75%), respectively. Since ASA operates based on the SSAM generated by the GAM module, we conducted this experiment to explore the synergistic effect between ASA and GAM. After further integrating ASA, the OA of the three datasets is further improved to 84.69 (+1.27%), 91.24 (+1.32%), and 73.43 (+1.52%), respectively, indicating a synergistic effect between the two components.

Notably, the performance improvement of GAM is more significant on the Berlin dataset (3.75%), which is higher than the 2.55% improvement on the other two datasets. This is due to the more scattered and interleaved distribution of land cover classes in the Berlin dataset, which significantly increases the difficulty of sample learning. GAM's gradient-guided masking strategy can accurately target regions with high reconstruction complexity (such as ambiguous boundaries between different land cover types), thereby more effectively optimizing the model's ability to learn discriminative features for hard samples.

In contrast, the performance improvement brought by ASA is more balanced across the three datasets (1.27%–1.52%). It dynamically adjusts attention weights through the semantic information of SSAM to enhance feature responses in key regions—after GAM determines "what to learn," ASA complements GAM by optimizing "how to learn." This confirms that GAM and ASA play different but complementary roles in improving model performance.

4) Analysis of Dynamic Fusion Weights α: We visualized fusion weights α [defined in (11)] across channels and pixels

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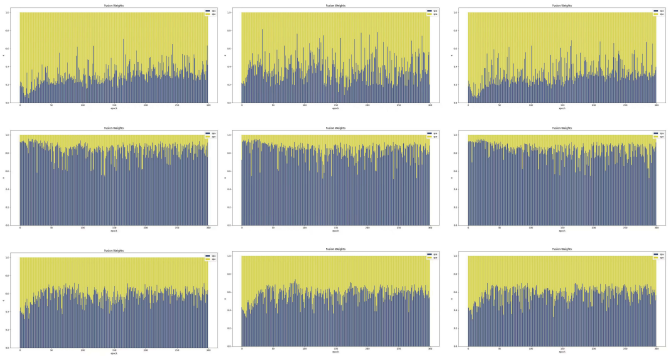

Fig. 14. Visualization of the fusion weights. Yellow represents spectral features, and blue represents spatial features.

(Fig. 14). These weights dynamically adjust with channels, spatial positions, and training progress, balancing contributions of spatial structural information and spectral discriminative features in real time. Ablation experiments on the Houston2013 dataset further show that introducing α improves the OA by 0.57%. Their contribution patterns fall into three categories: spatial-dominated, spectral-dominated, and synergistic. Notably, despite training fluctuations, the dominant mode for specific land cover categories remains stable—due to inherent properties of remote sensing data (e.g., buildings with distinct spatial morphology and vegetation with unique spectral curves have consistent modality preferences), which the model captures automatically without manual intervention.

5) Analysis of Computational Complexity: We evaluated the computational complexity of the models using the number of model parameters and floating-point operations (FLOPs) on the Augsburg dataset, with comparison results listed in Table IX. SimCLR, which employs a ResNet-50 deep network as its backbone, exhibits the largest parameter count (27.803 MB) and relatively high FLOPs due to its architectural depth. Supervised multimodal methods (e.g., MIFNet and NCGLF2) and contrastive self-supervised approaches (e.g., GraphGST, CMCL, and UACL) adopt lightweight designs to balance the processing of complex multimodal data and the avoidance of overfitting. In contrast, generative methods such as SS-MAE, LDS2AE, DAE-GSP, and the proposed method generally involve higher FLOPs, with the proposed method specifically having 663.75 FLOPs in nonupdating epochs and 1285.43 in updating epochs. If the proposed method's GAM module is replaced with standard random masking—thus removing the computational overhead of SSAM dynamic adjustments—its FLOPs can be reduced to 592.25. This arises from the necessity of integrating attention mechanisms and Transformer structures to handle the complexity of auxiliary tasks, even though LDS2AE utilizes significantly fewer parameters. Addressing this computational efficiency challenge represents an important direction for future research.

V. Conclusion

In conclusion, this study effectively addresses the limitations of traditional random masking strategies and attention mechanisms in remote sensing self-supervised learning by proposing the GAM method and ASA, constructing the GA-MAE framework. It significantly improves remote sensing image classification performance, with experiments on three public datasets verifying its effectiveness. It should be noted that GA-MAE has limitations: as a generative self-supervised learning method, its FLOPs are relatively high (e.g., 663.75 MB on the Augsburg dataset, higher than methods like CMCL and UACL) due to the introduction of GAM, ASA, and Transformer structures. This may increase training and inference time when processing large-scale datasets, limiting deployment in resource-constrained scenarios. Future work can explore lightweight Transformer architectures, hybrid CNN–Transformer structures, or introduce knowledge distillation to reduce parameter scale and FLOPs while maintaining feature learning capabilities, improving practical deployment efficiency. This research provides a novel and effective approach for remote sensing image classification, offering valuable ideas for future research in this field.

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